Lopos

More animal positions through dense vegetation

LoPoS tracking system · week-over-week field result · 2026-06-11

The setting

This system locates free-ranging animals in a natural, **vegetated environment**. Dense vegetation — plants, trees and canopy — is the main obstacle for the radio links: foliage absorbs and scatters the signals and repeatedly breaks the line-of-sight between the animal-borne tags and the receivers, and it keeps changing as plants grow and move. Everything below is about getting more reliable positions through that vegetation.

How the system works

- A **base station (the "sink")** coordinates everything over a long-range, low-frequency radio link: it tells each device when to act and collects all their reports.
- **Anchors** are fixed reference stations. Within each cell the central anchor keeps the others time-synchronised over ultra-wideband (UWB) radio; together they listen for the brief pulses sent by tags.
- **Tags** are carried by the animals and run on small batteries. A tag simply emits one UWB pulse in its assigned time slot. Its position is found from the tiny differences in *when* that single pulse reaches the surrounding anchors. A group of anchors that hears a tag together is a **cell**.

Headline

+57%	+149%	+193%	+8%	+76%
GRP1 positions / tag	GRP2 positions / tag	GRP3 positions / tag	GRP4 positions / tag	TOTAL positions vs baseline

"Positions per tag" = successful localizations per animal per minute. Higher is better; every group improved.

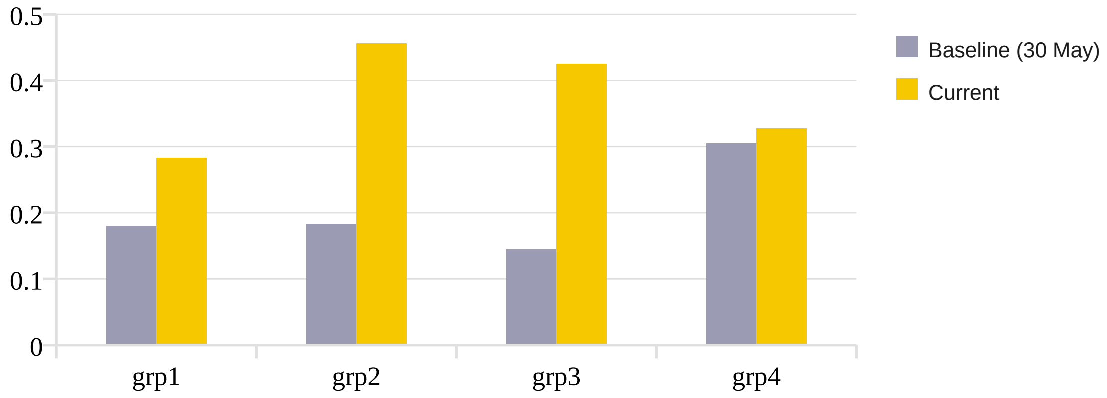
What we compared

Window	Dates (local)	System state
Baseline	Sat 30 May 11:00 → Sun 31 May ~12:00	Earlier system — original cell layout, simpler scheduling
Current	Wed 10 Jun 11:00 → Thu 11 Jun ~12:00	After the rebuilt cell network + improved scheduling went live

Matched ~25-hour windows (same start clock, same season, same animals — ~17 active tags per group). A "group" is one of the four zones of the study area.

Result, by zone

Zone	Baseline positions	Current positions	Per-tag rate	Change
grp1	4,617	7,262	0.18 → 0.28	▲ +57%
grp2	4,707	10,345	0.18 → 0.46	▲ +149%
grp3	3,516	10,283	0.14 → 0.42	▲ +193%
grp4	7,833	8,422	0.30 → 0.33	▲ +8%
TOTAL	**20,673**	**36,312**		▲ +76%

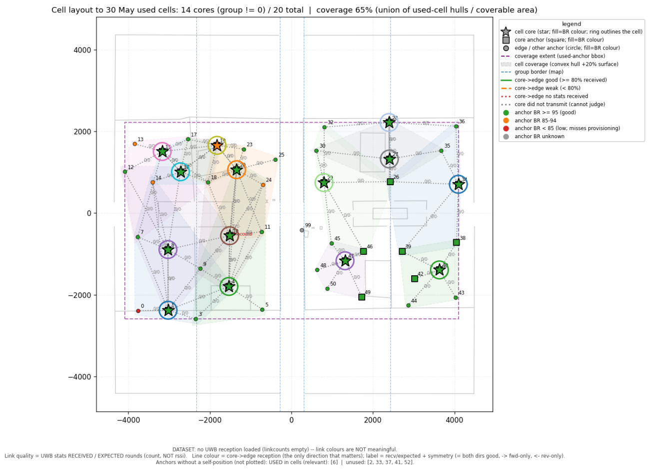


Positions per tag per minute, by zone.

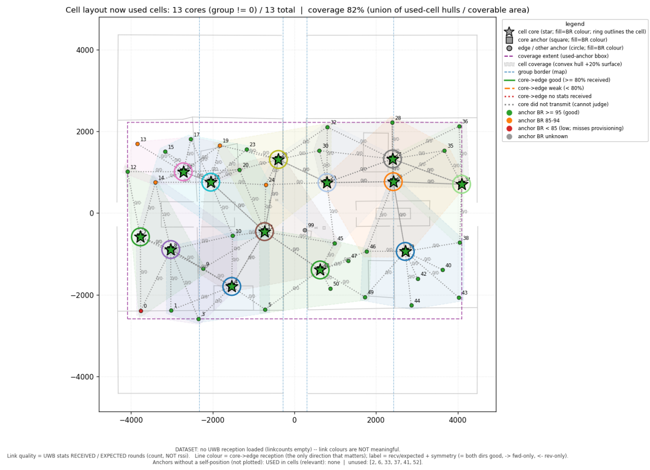
A rebuilt receiver network

We re-derived the cells from **measured** radio reception. Each animal is now covered by more, better-placed reference anchors. Green lines mark strong, reliable links between anchors.

Cell network — to 30 May



Cell network — now



Dots = anchors; stars = the central anchor of each cell; shaded areas = the ground each cell covers — covering more of the study area, with no idle, unused cells left over.

What we changed, and why it helps in vegetation

- **Rebuilt the cell network from real reception.** More anchors now hear each animal together, so a position can still be computed when foliage blocks some of the links — better coverage where the vegetation is densest.
- **A moving measurement window.** Because vegetation can block the scheduling messages, a tag sometimes acts on out-of-date instructions and transmits when the anchors are not listening for it. By letting the measurement window drift through the cycle, such a stray transmission now falls on quiet time instead of spoiling a live measurement — fewer wasted attempts.
- **Smarter, location-aware scheduling.** Each animal is measured in the cell nearest to where it actually is, instead of cycling blindly through all cells — more of every transmission turns into a usable position.
- **Transmissions matched to need.** Tags now transmit at their planned rate rather than over-transmitting, so the gains above come *without* extra radio traffic — extending tag battery life and study duration.

Net effect: the system delivers more reliable animal positions through dense vegetation, while transmitting no more often than before. Every zone improved; the hardest-to-cover zones (grp2, grp3) improved most.

Project work delivered

- **Code consolidation.** Recent improvements from the Wish deployment were integrated into the Ilvo code tree and merged into a single shared repository (support context).
- **Weekly data archiving.** Data is archived weekly, so each run processes only records that are at least one week old. Restricting the logic to a single weekly data set minimises the volume evaluated per run, keeping performance and scalability stable as the overall data volume grows.
- **Cell-network analysis.** Current and newly proposed cells were analysed from measured reception. Coverage improved from **20 cells / 68%** to **13 cells / 82%**.
- **Roaming cell selection.** Tag-to-cell assignment moved from round-robin to **roaming** — each animal is measured in the cell nearest its recent position.
- **Field follow-up.** Data follow-up runs **3 June – 13 July**.

Cell analysis (current vs proposed): shared Google Drive folder